

SAT Math Mastery · Week 8A**Angles, Similarity & Right-Triangle Trig — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** parallel lines & angles · triangle sum, exterior angle & triangle inequality · similar triangles & scaling of lengths/areas · Pythagorean triples & special right triangles · sine, cosine, tangent · $\sin x^\circ = \cos(90 - x)^\circ$ · triangle area**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Use vertical, linear-pair, corresponding, alternate-interior and co-interior angle relations, including with algebraic angle measures.
- S2.** Use the triangle angle sum, the exterior-angle theorem, and the triangle inequality (which side lengths can or cannot form a triangle).
- S3.** Prove or use similarity (AA, parallel line cutting a triangle) and set up proportions between corresponding sides, altitudes and perimeters.
- S4.** Scale lengths by k , areas by k^2 (and volumes by k^3); convert between a scale ratio and an area ratio or a percent change.
- S5.** Apply the Pythagorean theorem, Pythagorean triples, and the 45° - 45° - 90° and 30° - 60° - 90° side ratios.
- S6.** Write $\sin$, $\cos$, $\tan$ as ratios of sides; solve for a side from an angle (calculator in degree mode).
- S7.** Use the complementary-angle identity $\sin x^\circ = \cos(90 - x)^\circ$ and the fact that trig ratios depend only on the angle, not on the size of the triangle.
- S8.** Compute triangle area from base and height or from two sides and the included angle, and turn area information into a missing length or angle.

2 · REQUIRED FACTS

- Two intersecting lines make vertical angles (equal) and linear pairs (sum 180°). Parallel lines cut by a transversal give eight angles, each equal to θ or to $180^\circ - \theta$.
- Corresponding and alternate-interior angles are equal; co-interior (same-side interior) angles are supplementary.
- The angles of a triangle sum to 180° ; an exterior angle equals the sum of the two remote interior angles. The angles of an n -gon sum to $(n - 2)180^\circ$.
- Triangle inequality: each side is less than the sum of the other two, and greater than their difference. Equality means the triangle is degenerate (a flat segment).
- A line parallel to one side of a triangle cuts the other two sides proportionally and creates a smaller triangle similar to the whole.
- Similarity ratio k : lengths (sides, altitudes, perimeters) scale by k , areas by k^2 . Angles never change.
- Common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41, and their multiples.
- In a right triangle the two acute angles are complementary, so $\sin A = \cos B$ and $\tan A \cdot \tan B = 1$ when $A + B = 90^\circ$.
- The altitude to the hypotenuse of a right triangle splits it into two triangles similar to the whole triangle and to each other.

3 · REQUIRED FORMULAS

$$\angle_{\text{ext}} = \angle_1 + \angle_2, \quad |a - b| < c < a + b, \quad a^2 + b^2 = c^2, \quad \frac{\text{new}}{\text{old}} = k \Rightarrow \frac{\text{area}_{\text{new}}}{\text{area}_{\text{old}}} = k^2$$

$$45^\circ\text{-}45^\circ\text{-}90^\circ : s, s, s\sqrt{2} \quad 30^\circ\text{-}60^\circ\text{-}90^\circ : s, s\sqrt{3}, 2s \quad \text{equilateral: } h = \frac{\sqrt{3}}{2}s, A = \frac{\sqrt{3}}{4}s^2$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}, \quad \sin x^\circ = \cos(90 - x)^\circ, \quad A = \frac{1}{2}bh = \frac{1}{2}ab \sin C$$

4 · MINI THEORY SHEET

Angle chasing in a figure. Mark every angle you can find in this order: (1) vertical angles and linear pairs, (2) parallel-line relations, (3) triangle sum / exterior angle. When angles carry algebra, decide first whether the two angles are *equal* or *supplementary*, then write one equation.

Which angle relation?

Position of the two angles	Relation
opposite each other at one crossing	equal (vertical)
same corner at two crossings (same side of the transversal)	equal (corresponding)
between the lines, opposite sides of the transversal	equal (alternate interior)
between the lines, same side of the transversal	sum 180° (co-interior)

Similar triangles. Match vertices by angle, write the proportion with matching sides in the same order, then cross-multiply. A segment parallel to a side gives $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ (part-to-whole), and also $\frac{AD}{DB} = \frac{AE}{EC}$ (part-to-part). Do not mix part-to-part with part-to-whole.

Scaling. If lengths scale by k , areas by k^2 . A “ $p\%$ larger area” means a factor $1 + \frac{p}{100}$ on area, so $k = \sqrt{1 + \frac{p}{100}}$.

Right triangles. First look for a triple or a special triangle before squaring anything. A leg is $\sqrt{c^2 - a^2}$; a diagonal of a rectangle is $\sqrt{\ell^2 + w^2}$.

Trigonometry. Label opposite, adjacent and hypotenuse *relative to the angle you are given*. Ratios are properties of the angle, so $\sin A$ is the same in every triangle similar to the one shown. If $\sin A = \cos B$ for acute angles A and B , then $A + B = 90^\circ$.

Triangle area. With two sides a, b and included angle C , $A = \frac{1}{2}ab \sin C$. Since $\sin C = \sin(180^\circ - C)$, an area condition often has *two* possible angles C and $180^\circ - C$.

Speed shortcuts.

Situation	Shortcut
third side of a triangle is a whole number	$ a - b < c < a + b$: count integers strictly between
angle expressions in x , same side between parallels	set the sum to 180° (not equal)
area ratio given, find a length	take the square root of the area ratio first
parallel segment, find a whole side	use $\frac{\text{part}}{\text{whole}}$ with the whole = part + rest
$\sin(\cdot) = \cos(\cdot)$ with unknowns	add the two angle expressions and set the sum to 90
leg and hypotenuse differ by 1	$a^2 = 2b + 1$, so $b = \frac{a^2 - 1}{2}$ when $c = b + 1$
right-triangle ratios in a bigger figure	find the similar small right triangle; the ratio is the same
$\tan A$ given	draw the triangle with legs opp, adj; hypotenuse by triple or Pythagoras

5 · GUIDED EXAMPLES

Example 1 — Parallel lines with algebraic angles

Lines $\ell \parallel m$ are cut by a transversal. Two angles lie *between* the lines on the *same side* of the transversal and measure $(2x + 15)^\circ$ and $(3x - 10)^\circ$. Find both angles.

Same-side interior angles are supplementary: $(2x + 15) + (3x - 10) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow x = 35$.

The angles are $2(35) + 15 = 85^\circ$ and $3(35) - 10 = 95^\circ$. Check: $85 + 95 = 180$.

Example 2 — Similar triangles from a parallel line

In $\triangle ABC$, D lies on $\overline{AB}$ and E on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. If $AD = 4$, $DB = 6$ and $DE = 5$, find BC and the ratio of the areas $[ADE] : [ABC]$.

$\triangle ADE \sim \triangle ABC$ with ratio $k = \frac{AD}{AB} = \frac{4}{4+6} = \frac{2}{5}$ (part over *whole*).

$BC = \frac{DE}{k} = \frac{5}{2/5} = 12.5$. Area ratio $= k^2 = \frac{4}{25}$.

Example 3 — Right-triangle trig and the cofunction identity

(a) In right $\triangle ABC$ with $\angle C = 90^\circ$, $AC = 8$ and $BC = 15$. Find $\sin A$ and $\cos B$.

(b) If $\sin(4x - 2)^\circ = \cos(x + 7)^\circ$ and both angles are acute, find x .

(a) $AB = \sqrt{8^2 + 15^2} = 17$. $\sin A = \frac{BC}{AB} = \frac{15}{17}$, and $\cos B = \frac{BC}{AB} = \frac{15}{17}$, the same value (complementary angles).

(b) Complementary angles: $(4x - 2) + (x + 7) = 90 \Rightarrow 5x + 5 = 90 \Rightarrow x = 17$.

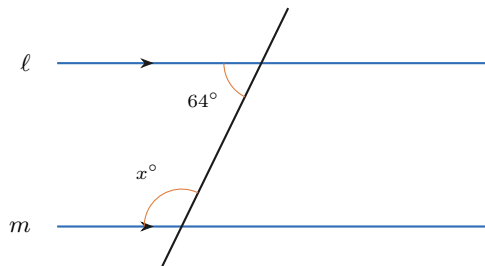
6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to line m . What is the value of x ?



(A) 64

(B) 116 ✓

(C) 26

(D) 154

Worked solution

1. The marked angle at the top and the angle x° lie between the parallel lines on the same side of the transversal (co-interior angles), so they are supplementary.

2. $x = 180 - 64 = 116$.

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

In triangle ABC , the measure of angle A is 38° . The exterior angle at vertex C , formed by extending side $\overline{BC}$ beyond C , measures 115° . What is the measure, in degrees, of angle B ?

Worked solution

1. An exterior angle equals the sum of the two remote interior angles: $115 = 38 + B$.

2. $B = 115 - 38 = 77$.

Answer: 77

3.

EASY · NO CALCULATOR NEEDED

The lengths of two sides of a triangle are 7 and 12. Which of the following could be the length of the third side?

(A) 20

(B) 5

(C) 10 ✓

(D) 19

Worked solution

1. The third side c must satisfy $12 - 7 < c < 12 + 7$, that is $5 < c < 19$.

2. Only 10 lies strictly between 5 and 19 (5 and 19 would give a flat triangle).

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

A rectangular banner has a width of 20 inches and a diagonal of 25 inches. What is the area, in square inches, of the banner?

Worked solution

1. The diagonal is the hypotenuse of a right triangle with legs 20 and the length ℓ : $\ell = \sqrt{25^2 - 20^2} = \sqrt{225} = 15$ (a 3-4-5 triple scaled by 5).

2. Area = $20 \cdot 15 = 300$.

Answer: 300

5.

EASY · NO CALCULATOR NEEDED

In right triangle ABC , angle C is the right angle, $AC = 15$ and $BC = 8$. What is the value of $\sin A$?

(A) $\frac{8}{17}$ ✓

(B) $\frac{15}{17}$

(C) $\frac{8}{15}$

(D) $\frac{17}{8}$

Worked solution

1. $AB = \sqrt{15^2 + 8^2} = \sqrt{289} = 17$.

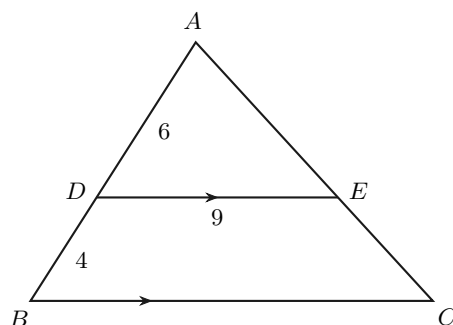
2. The side opposite angle A is $BC = 8$, so $\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{8}{17}$.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

In the figure, $\overline{DE}$ is parallel to $\overline{BC}$. What is the length of $\overline{BC}$?



(A) 22.5

(B) 6

(C) 13.5

(D) 15 ✓

Worked solution

1. Since $\overline{DE} \parallel \overline{BC}$, $\triangle ADE \sim \triangle ABC$.

2. $AB = AD + DB = 6 + 4 = 10$, so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{6}{10}$.

3. $BC = \frac{9 \cdot 10}{6} = 15$.

Answer: (D)

7.

MEDIUM · NO CALCULATOR NEEDED

Triangle ABC is similar to triangle DEF , with $AB = 8$ and the corresponding side $DE = 12$. The area of triangle ABC is 40 square units. What is the area, in square units, of triangle DEF ?

Worked solution

1. The scale factor from ABC to DEF is $k = \frac{12}{8} = \frac{3}{2}$.

2. Areas scale by $k^2 = \frac{9}{4}$.

3. Area of $DEF = 40 \cdot \frac{9}{4} = 90$.

Answer: 90

8. MEDIUM · NO CALCULATOR NEEDED
The measures of the three angles of a triangle are $(3x + 10)^\circ$, $(5x - 14)^\circ$ and $(x + 4)^\circ$. What is the measure, in degrees, of the largest angle?

- (A) 116 (B) 70
(C) 86 ✓ (D) 100

Worked solution

1. Angle sum: $(3x + 10) + (5x - 14) + (x + 4) = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$.
2. The angles are 70° , 86° and 24° (sum 180°).
3. The largest is 86° .

Answer: (C)

9. MEDIUM · NO CALCULATOR NEEDED
The diagonal of a square has length $10\sqrt{2}$. What is the perimeter of the square?

- (A) 20 (B) 40 ✓
(C) $40\sqrt{2}$ (D) 100

Worked solution

1. A square's diagonal splits it into two 45° - 45° - 90° triangles, so the diagonal is $s\sqrt{2}$.
2. $s\sqrt{2} = 10\sqrt{2} \Rightarrow s = 10$.
3. Perimeter $= 4s = 40$.

Answer: (B)

10. MEDIUM · NO CALCULATOR NEEDED
The altitude of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?

Worked solution

1. The altitude splits the triangle into two 30° - 60° - 90° triangles; the altitude is the long leg, $\frac{\sqrt{3}}{2}s$.
2. $\frac{\sqrt{3}}{2}s = 6\sqrt{3} \Rightarrow s = 12$.
3. Perimeter $= 3 \cdot 12 = 36$.

Answer: 36

11. MEDIUM · NO CALCULATOR NEEDED
In right triangle ABC , angle C is the right angle and $\tan A = \frac{5}{12}$. What is the value of $\cos B$?

- (A) $\frac{12}{13}$ (B) $\frac{5}{12}$
(C) $\frac{13}{5}$ (D) $\frac{5}{13}$ ✓

Worked solution

1. Take the legs as $BC = 5$ (opposite A) and $AC = 12$ (adjacent to A); then $AB = \sqrt{5^2 + 12^2} = 13$.
2. $\cos B = \frac{\text{adj to } B}{\text{hyp}} = \frac{BC}{AB} = \frac{5}{13}$. (Equivalently $\cos B = \sin A$ since $A + B = 90^\circ$.)

Answer: (D)

12.

MEDIUM · CALCULATOR

A wheelchair ramp makes an angle of 12° with the level ground and has a horizontal run of 15 meters. To the nearest tenth of a meter, how high does the ramp rise?

Worked solution

1. The rise is opposite the 12° angle and the run is adjacent: $\tan 12^\circ = \frac{\text{rise}}{15}$.
 2. $\text{rise} = 15 \tan 12^\circ \approx 15(0.2126) \approx 3.19$ (calculator in degree mode).
 3. To the nearest tenth: 3.2.
- Answer:** 3.2

13.

MEDIUM · NO CALCULATOR NEEDED

On a floor plan, 1 centimeter represents 2 meters. A rectangular room has an area of 18 square centimeters on the plan. What is the actual area of the room, in square meters?

- (A) 72 ✓ (B) 144
(C) 9 (D) 36

Worked solution

1. Lengths scale by 2 (cm $\rightarrow$ m), so areas scale by $2^2 = 4$.
 2. Actual area = $18 \cdot 4 = 72$ square meters.
- Answer:** (A)

14.

HARD · NO CALCULATOR NEEDED

A triangle has side lengths 8, 15 and x , where $x > 0$. For which of the following values of x is the triangle a right triangle?

I. $x = 17$ II. $x = \sqrt{161}$ III. $x = 23$

- (A) II only (B) I and II only ✓
(C) I, II, and III (D) I only

Worked solution

1. If x is the hypotenuse: $x^2 = 8^2 + 15^2 = 289 \Rightarrow x = 17$. So I works.
 2. If 15 is the hypotenuse: $x^2 = 15^2 - 8^2 = 161 \Rightarrow x = \sqrt{161}$. So II works.
 3. $x = 23$ gives $8 + 15 = 23$, which violates the triangle inequality (a flat segment), so III does not form a triangle at all.
 4. I and II only.
- Answer:** (B)

15.

HARD · NO CALCULATOR NEEDED

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. If $AD = x$, $DB = x + 2$, $AE = 6$, $EC = 8$ and $DE = 15$, what is the length of $\overline{BC}$?

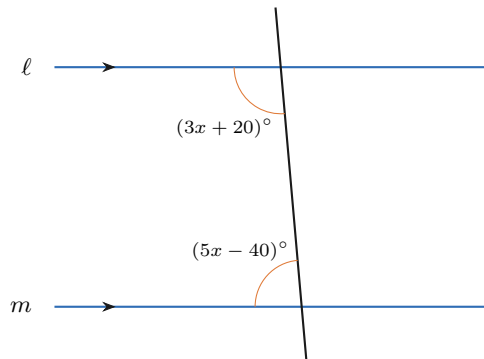
Worked solution

1. $\overline{DE} \parallel \overline{BC}$ gives $\frac{AD}{DB} = \frac{AE}{EC}$: $\frac{x}{x+2} = \frac{6}{8} = \frac{3}{4}$.
 2. $4x = 3x + 6 \Rightarrow x = 6$, so $AD = 6$, $DB = 8$ and $AB = 14$.
 3. $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow BC = \frac{15 \cdot 14}{6} = 35$.
- Answer:** 35

16.

HARD · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to line m . What is the measure, in degrees, of the angle labeled $(5x - 40)^\circ$?



(A) 110

(B) 25

(C) 85 ✓

(D) 95

Worked solution

1. The two labeled angles are between the parallel lines on the same side of the transversal, so they are supplementary.
2. $(3x + 20) + (5x - 40) = 180 \Rightarrow 8x - 20 = 180 \Rightarrow x = 25$.
3. $5x - 40 = 5(25) - 40 = 85$. (Check: the other angle is $3(25) + 20 = 95$, and $85 + 95 = 180$.)

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

Two parallel lines are cut by a transversal, forming eight angles. One of the eight angles measures 50° . Which of the following could *NOT* be the sum of the measures of some of these eight angles (each angle may be used at most once)?

(A) 500 ✓

(B) 280

(C) 360

(D) 410

Worked solution

1. Each of the eight angles is 50° or 130° : four of each. Any sum has the form $50k + 130m$ with $0 \leq k \leq 4$ and $0 \leq m \leq 4$.
2. $280 = 50(3) + 130(1)$; $360 = 50(2) + 130(2)$; $410 = 50(3) + 130(2)$.
3. 500: for $m = 0, 1, 2, 3, 4$ the remainder $500 - 130m$ is 500, 370, 240, 110, -20. A valid k needs the remainder to equal $50k$ with $0 \leq k \leq 4$ (so at most 200); only 500, 370, 240, 110 are candidates and none is both a multiple of 50 and at most 200. So 500 is impossible.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

A triangle has side lengths 6, 11 and k , where k is an integer. If the triangle is *not* isosceles, how many different values of k are possible?

(A) 10

(B) 11

(C) 12

(D) 9 ✓

Worked solution

1. Triangle inequality: $11 - 6 < k < 11 + 6$, so $5 < k < 17$ and $k \in \{6, 7, \dots, 16\}$, which is 11 values.
2. Isosceles happens when $k = 6$ or $k = 11$; remove both: $11 - 2 = 9$.

Answer: (D)

19.

HARD · NO CALCULATOR NEEDED

A right triangle has integer side lengths. Its shorter leg is 11, and its hypotenuse is exactly 1 more than its longer leg. What is the perimeter of the triangle?

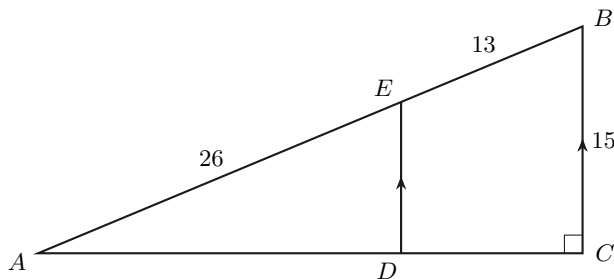
Worked solution

1. Let the longer leg be b ; the hypotenuse is $b + 1$. Then $11^2 + b^2 = (b + 1)^2$.
 2. $121 + b^2 = b^2 + 2b + 1 \Rightarrow 2b = 120 \Rightarrow b = 60$, hypotenuse 61 (the triple 11-60-61).
 3. Perimeter = $11 + 60 + 61 = 132$.
- Answer:** 132

20.

HARD · NO CALCULATOR NEEDED

In the figure, $\angle C$ is a right angle and $\overline{DE}$ is parallel to $\overline{BC}$. What is the value of $\cos \angle AED$?



- (A) $\frac{5}{12}$
 (C) $\frac{5}{13}$ ✓

- (B) $\frac{2}{3}$
 (D) $\frac{12}{13}$

Worked solution

1. Because $\overline{DE} \parallel \overline{BC}$, $\angle AED = \angle ABC$ (corresponding angles), so $\cos \angle AED = \cos B$.
 2. $AB = AE + EB = 26 + 13 = 39$.
 3. In right triangle ABC : $\cos B = \frac{BC}{AB} = \frac{15}{39} = \frac{5}{13}$.
- Answer:** (C)

21.

HARD · NO CALCULATOR NEEDED

If $\sin(2x - 6)^\circ = \cos(3x + 16)^\circ$, where $(2x - 6)^\circ$ and $(3x + 16)^\circ$ are both acute angles, what is the value of x ?

Worked solution

1. Sine and cosine are equal for complementary angles: $(2x - 6) + (3x + 16) = 90$.
 2. $5x + 10 = 90 \Rightarrow x = 16$. (Check: 26° and 64° are complementary.)
- Answer:** 16

22.

HARD · NO CALCULATOR NEEDED

The table gives information about two similar triangular plots of land, X and Y .

Plot	Side lengths (m)	Area (m^2)
X	9, 12, 15	—
Y	—	384

What is the perimeter, in meters, of plot Y ?

(A) 32

(B) 96 ✓

(C) 256

(D) 288

Worked solution

1. X is a right triangle (9-12-15): area = $\frac{1}{2}(9)(12) = 54$, perimeter = 36.

2. Area ratio = $\frac{384}{54} = \frac{64}{9}$, so the scale factor is $k = \sqrt{\frac{64}{9}} = \frac{8}{3}$.

3. Perimeter of $Y = 36 \cdot \frac{8}{3} = 96$.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

Two sides of a triangle have lengths 8 and 10. The area of the triangle is 20. The angle between these two sides has a measure of C° , where $0 < C < 180$. What is the sum of all possible values of C ?

Worked solution

1. Area = $\frac{1}{2}(8)(10) \sin C = 40 \sin C = 20 \Rightarrow \sin C = \frac{1}{2}$.

2. In $(0^\circ, 180^\circ)$, $\sin C = \frac{1}{2}$ has two solutions: $C = 30^\circ$ and $C = 150^\circ$.

3. Sum = $30 + 150 = 180$.

Answer: 180

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , $AB = 20$ and $BC = 12$. Point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$, and the area of triangle ADE equals the area of quadrilateral $DBCE$. What is the value of $(AD)(DE)$?

Worked solution

1. $\triangle ADE \sim \triangle ABC$ with ratio $k = \frac{AD}{AB}$. Equal areas means $[ADE] = \frac{1}{2}[ABC]$.

2. Areas scale by k^2 : $k^2 = \frac{1}{2}$, so $k = \frac{1}{\sqrt{2}}$.

3. $AD = 20k = \frac{20}{\sqrt{2}} = 10\sqrt{2}$ and $DE = 12k = \frac{12}{\sqrt{2}} = 6\sqrt{2}$.

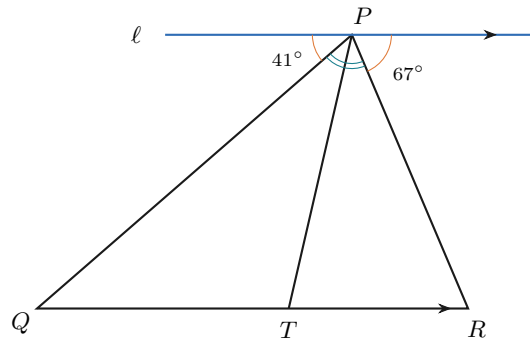
4. $(AD)(DE) = (10\sqrt{2})(6\sqrt{2}) = 60 \cdot 2 = 120$.

Answer: 120

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to side $\overline{QR}$ of triangle PQR . Point T lies on $\overline{QR}$, and $\overline{PT}$ bisects $\angle QPR$. What is the measure, in degrees, of $\angle PTR$?



(A) 72

(B) 36

(C) 103

(D) 77 ✓

Worked solution

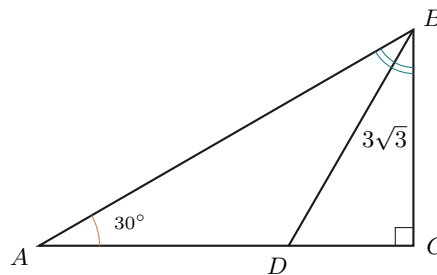
1. Alternate interior angles with $\ell \parallel \overline{QR}$: $\angle Q = 41^\circ$ and $\angle R = 67^\circ$.
2. Triangle sum: $\angle QPR = 180 - 41 - 67 = 72^\circ$; the bisector gives $\angle QPT = \angle TPR = 36^\circ$.
3. $\angle PTR$ is an exterior angle of $\triangle PQT$, so $\angle PTR = \angle Q + \angle QPT = 41 + 36 = 77$.
4. Check in $\triangle PTR$: $\angle PTR = 180 - 36 - 67 = 77$.

Answer: (D)

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $BC = 3\sqrt{3}$, and $\overline{BD}$ bisects $\angle ABC$. What is the length of $\overline{AD}$?



Worked solution

1. $\angle ABC = 90^\circ - 30^\circ = 60^\circ$, so $\angle DBC = 30^\circ$.
2. In $\triangle ABC$ (30° - 60° - 90°), BC is the short leg, so $AC = BC\sqrt{3} = 3\sqrt{3} \cdot \sqrt{3} = 9$.
3. In $\triangle BCD$ (30° - 60° - 90°), BC is the long leg, so $CD = \frac{BC}{\sqrt{3}} = 3$.
4. $AD = AC - CD = 9 - 3 = 6$. (Check: $\angle ABD = \angle A = 30^\circ$, so $AD = BD = 2CD = 6$.)

Answer: 6

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , angle C is the right angle. For a constant x , $\cos A = \frac{2x+4}{13}$ and $\sin B = \frac{5x-8}{13}$. What is the value of $\tan A$?

(A) $\frac{5}{12}$ ✓

(B) $\frac{12}{5}$

(C) $\frac{5}{13}$

(D) $\frac{12}{13}$

Worked solution

1. A and B are the acute angles of a right triangle, so they are complementary and $\cos A = \sin B$.

2. $2x + 4 = 5x - 8 \Rightarrow 3x = 12 \Rightarrow x = 4$, so $\cos A = \frac{12}{13}$.

3. Adjacent = 12, hypotenuse = 13, so opposite = $\sqrt{13^2 - 12^2} = 5$ (the triple 5-12-13).

4. $\tan A = \frac{\text{opp}}{\text{adj}} = \frac{5}{12}$.

Answer: (A)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

How many different triangles have integer side lengths and a longest side of exactly 10? (Triangles with the same three side lengths are counted once.)

Worked solution

1. Write the sides as $a \leq b \leq 10$. The triangle inequality (longest side < sum of the others) requires $a + b > 10$.

2. For each b , the allowed a run from $11 - b$ to b , giving $2b - 10$ values: $b = 6, 7, 8, 9, 10$ give 2, 4, 6, 8, 10.

3. Total = $2 + 4 + 6 + 8 + 10 = 30$.

Answer: 30

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

Triangle Q is similar to triangle P , and the area of Q is 125% greater than the area of P . The sum of the lengths of an altitude of P and the corresponding altitude of Q is 15 centimeters. What is the length, in centimeters, of the altitude of Q ?

Worked solution

1. “125% greater” means the area of Q is 2.25 times the area of P .

2. Areas scale by k^2 : $k^2 = 2.25 \Rightarrow k = 1.5$, so $\text{altitude}_Q = 1.5 \cdot \text{altitude}_P$.

3. $\text{altitude}_P + 1.5 \text{altitude}_P = 15 \Rightarrow \text{altitude}_P = 6$.

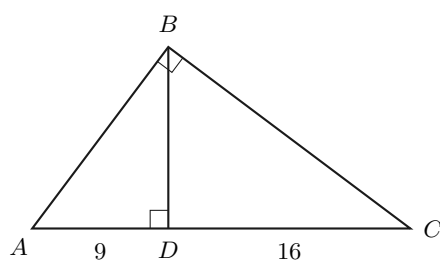
4. $\text{altitude}_Q = 1.5 \cdot 6 = 9$.

Answer: 9

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\angle ABC$ is a right angle and $\overline{BD}$ is perpendicular to $\overline{AC}$. What is the value of $\tan \angle ABD$?



(A) $\frac{9}{16}$

(B) $\frac{3}{4}$ ✓

(C) $\frac{4}{3}$

(D) $\frac{3}{5}$

Worked solution

1. In right $\triangle ABC$ with altitude $\overline{BD}$ to the hypotenuse, $BD^2 = AD \cdot DC = 9 \cdot 16 = 144$, so $BD = 12$.

2. In right $\triangle ABD$, the angle at B has opposite side $AD = 9$ and adjacent side $BD = 12$.

3. $\tan \angle ABD = \frac{9}{12} = \frac{3}{4}$. (Check: $\angle ABD = \angle C$, and $\tan C = \frac{AB}{BC} = \frac{15}{20} = \frac{3}{4}$.)

Answer: (B)